

SingleCell – Código completo del pipeline

Todos los archivos del workflow, por bloques

Pipeline SingleCell

2026-06-24

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1 Resumen

Este documento contiene el código fuente **completo** de todos los archivos del repositorio `SingleCell/workflow/` – sin recortar ni resumir – organizado en bloques, uno por archivo. Para una explicación paso a paso con resultados e imágenes, ver el reporte `capitulo3_results_report.pdf`. Para la referencia de funciones con parámetros y ejemplos cortos, ver `README.md`.

2 load_libraries.R

Carga de paquetes R (capítulos 1 y 2).

[illegible]

3 custom_seurat.R

Utilidades de plotting custom para Seurat (gráficos de barras de cluster, etc.).

```
\CommentTok{\# =====}
\CommentTok{\# custom\_seurat.R {-}{-} Custom Seurat plot utilities}
\CommentTok{\# =====}

\CommentTok{\#\textquotesingle{} Cell Count Bar Chart per Cluster and Sample}
\CommentTok{\#\textquotesingle{}}
\CommentTok{\#\textquotesingle{} Produces a two{-}panel figure: (left) fraction of cells per dataset within}
\CommentTok{\#\textquotesingle{} each identity/cluster; (right) total cell count per cluster on a log10 scale}
\CommentTok{\#\textquotesingle{} with count labels.}
\CommentTok{\#\textquotesingle{}}
\CommentTok{\#\textquotesingle{} @param srat Seurat object. Uses active Idents() as the grouping variable.}
\CommentTok{\#\textquotesingle{} @return A patchwork ggplot object.}
\CommentTok{\#\textquotesingle{} @export}
\NormalTok{plot\_integrated\_clusters }\OtherTok{\textless{}{-}} \ControlFlowTok{function}\NormalTok{(srat) \{}

\NormalTok{  count\_table }\OtherTok{\textless{}{-}} \FunctionTok{table}\NormalTok{ }\FunctionTok{Idents}\NormalTok{(srat) }\FunctionTok{as.data.frame.matrix}\NormalTok{(count\_table)}
\NormalTok{  count\_mtx }\OtherTok{\textless{}{-}} \FunctionTok{as.data.frame.matrix}\NormalTok{(count\_table)}
\NormalTok{  count\_mtx }\SpecialCharTok{$}\NormalTok{identity }\OtherTok{\textless{}{-}} \FunctionTok{rownames}\NormalTok{(count\_mtx)}
\NormalTok{  melt\_mtx }\OtherTok{\textless{}{-}} \NormalTok{count\_mtx }\FunctionTok{reshape2}\NormalTok{ }\FunctionTok{melt}\NormalTok{(count\_mtx)}
\NormalTok{  melt\_mtx }\SpecialCharTok{$}\NormalTok{identity }\OtherTok{\textless{}{-}} \FunctionTok{as.factor}\NormalTok{(melt\_mtx$identity)}

\NormalTok{  cluster\_size }\OtherTok{\textless{}{-}} \FunctionTok{aggregate}\NormalTok{(count\_mtx[,1], melt\_mtx$identity, FUN = sum)}
\NormalTok{  sorted\_labels }\OtherTok{\textless{}{-}} \NormalTok{cluster\_size }\FunctionTok{sort}\NormalTok{(cluster\_size)}
\NormalTok{  cluster\_size }\SpecialCharTok{$}\NormalTok{identity }\OtherTok{\textless{}{-}} \FunctionTok{factor}\NormalTok{(cluster\_size)}
\NormalTok{  melt\_mtx }\SpecialCharTok{$}\NormalTok{identity }\OtherTok{\textless{}{-}} \FunctionTok{factor}\NormalTok{(melt\_mtx$identity)}

  \FunctionTok{colnames}\NormalTok{(melt\_mtx) }\FunctionTok{setNames}\NormalTok{(melt\_mtx[,1], melt\_mtx$identity)}
\NormalTok{  p\_fraction }\OtherTok{\textless{}{-}} \FunctionTok{ggplot}\NormalTok{(melt\_mtx, aes(x = identity, y = fraction))}
\NormalTok{  p\_fraction }\FunctionTok{geom\_bar}\NormalTok{(fill = "#f0f0f0", color = "black", width = 1)}
\NormalTok{  p\_fraction }\FunctionTok{scale\_x}\NormalTok{(breaks = Idents(srat), labels = Idents(srat))}
\NormalTok{  p\_fraction }\FunctionTok{scale\_y}\NormalTok{(breaks = c(0, 0.2, 0.4, 0.6, 0.8, 1))}
\NormalTok{  p\_fraction }\FunctionTok{labs}\NormalTok{(x = "Identity", y = "Fraction of cells per dataset")}
\NormalTok{  p\_fraction }\FunctionTok{theme}\NormalTok{(base = theme_minimal(), legend.position = "top")}

\NormalTok{  p\_count }\OtherTok{\textless{}{-}} \FunctionTok{ggplot}\NormalTok{(cluster\_size, aes(x = identity, y = count))}
\NormalTok{  p\_count }\FunctionTok{geom\_bar}\NormalTok{(fill = "#f0f0f0", color = "black", width = 1)}
\NormalTok{  p\_count }\FunctionTok{scale\_x}\NormalTok{(breaks = Idents(srat), labels = Idents(srat))}
\NormalTok{  p\_count }\FunctionTok{scale\_y}\NormalTok{(breaks = c(0, 1000, 2000, 3000, 4000, 5000))}
\NormalTok{  p\_count }\FunctionTok{labs}\NormalTok{(x = "Identity", y = "Cells per identity (log10 scale)")}
\NormalTok{  p\_count }\FunctionTok{theme}\NormalTok{(base = theme_minimal(), legend.position = "top")}

\NormalTok{  p\_fraction }\SpecialCharTok{+}\NormalTok{p\_count }\FunctionTok{plot\_layout}\NormalTok{(p\_fraction, p\_count)}
\NormalTok{}
```

4 **ScRNA_Analysis_Functions.R**

Funciones núcleo de capítulos 1 y 2: QC, doublets, pseudobulk, anotación, DESeq2, volcano, heatmap, GO enrichment, hdWGCNA, save helpers. Archivo más grande del pipeline (5046 líneas) – firmas y ejemplos resumidos también en README.md.

5 capitulo1_single_cell.R

Script principal de capítulo 1: QC -> filtrado -> integración Harmony -> clustering -> anotación -> curación -> exportación a h5ad.

```
\CommentTok{\# =====}
\CommentTok{\# capitulo1\_single\_cell.R {-}{-} Single{-}Cell RNA{-}seq Analysis (Part 1)}
\CommentTok{\# =====}
\CommentTok{\# Covers QC, integration, clustering, annotation, and export to h5ad.}
\CommentTok{\# Output: pbmc\_harmony\_curated.rds (used by capitulo2\_pseudobulk\_de.R)}
\CommentTok{\#          pbmc\_harmony\_curated.h5ad (used by capitulo3\_pseudotime.ipynb)}
\CommentTok{\# =====}

\CommentTok{\# =====}
\CommentTok{\# Single{-}Cell RNA{-}seq Analysis Pipeline}
\CommentTok{\# =====}
\CommentTok{\# Protocol: Methods in Molecular Biology}
\CommentTok{\# }
\CommentTok{\# Description:}
\CommentTok{\#   End{-}to{-}end scRNA{-}seq pipeline covering:}
\CommentTok{\#     Part 1 {-}{-} QC, integration, clustering, annotation, and export}
\CommentTok{\#     Part 2 {-}{-} Pseudobulk differential expression and GO enrichment}
\CommentTok{\# }
\CommentTok{\# Helper scripts (fully documented at https://github.com/cliford2001/ScRNASeq{-}Docker):}
\CommentTok{\#   load\_libraries.R           {-}{-} loads all required R packages}
\CommentTok{\#   ScRNA\_Analysis\_Functions.R {-}{-} core functions (QC, clustering, annotation, DE, GO)}
\CommentTok{\#   custom\_seurat.R            {-}{-} custom Seurat plot utilities (cluster bar charts)}
\CommentTok{\# }
\CommentTok{\# Usage:}
\CommentTok{\#   Run sections sequentially. Section 12 (Cell{-}Type Curation) must be}
\CommentTok{\#   run interactively {-}{-} do NOT source the entire script with it active.}
\CommentTok{\# }
\CommentTok{\# Organism:}
\CommentTok{\#   Default: Arabidopsis thaliana. Adapting to another organism requires}
\CommentTok{\#   updating only the organelle patterns, GO database, and key type}
\CommentTok{\#   (each marked clearly in the relevant section below).}
\CommentTok{\# }
\CommentTok{\# Reproducibility:}
\CommentTok{\#   Fixed random seed (1807). Intermediate results written to disk.}
\CommentTok{\# =====}

\CommentTok{\# =====}
\CommentTok{\# CONFIGURATION {-}{-} Edit only this block before running the pipeline}
\CommentTok{\# =====}


\CommentTok{\# Directory containing the pipeline helper scripts.}
\CommentTok{\# Inside the Docker container this is typically /workspace/ScRNASeq{-}Docker}
\normaltok{PIPELINE\_DIR }\othertok{\textless{}\{-}\} \StringTok{"workspace/ScRNASeq{-}Docker/workflow"}


\CommentTok{\# Root directory for your project data and results.}
\CommentTok{\# All result files will be written to DATA\_DIR/resultados/\textless{}\step\textgreater{}\{/}
\normaltok{DATA\_DIR    }\othertok{\textless{}\{-}\} \StringTok{"workspace/."}
\normaltok{base\_dir    }\othertok{\textless{}\{-}\} \FunctionTok{file.path}\normaltok{(DATA\_DIR, )\StringTok{"resultado"}}


\CommentTok{\# {-}{-} Sample manifest (CellRanger filtered\_feature\_bc\_matrix) {-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}}
\CommentTok{\# Add one entry per sample. Each entry needs:}
\CommentTok{\#   file      {-}{-} path to the filtered\_feature\_bc\_matrix/ directory (relative to DATA\_DIR)}
\CommentTok{\#   label     {-}{-} unique name for this sample (appears in all plots)}
\CommentTok{\#   condition {-}{-} experimental group this sample belongs to}
\normaltok{samples }\othertok{\textless{}\{-}\} \FunctionTok{list}\normaltok{{}
  \FunctionTok{list}\normaltok{{}\Attributetok{file =} \StringTok{"cellranger/Sample\_ON outs/filtered\_feature\_bc\_m
  \FunctionTok{list}\normaltok{{}\Attributetok{file =} \StringTok{"cellranger/Sample\_O5N outs/filtered\_feature\_bc\_m
```

[illegible]

[illegible]


```
\CommentTok{\# Checkpoint {-}{-} restore with: pbmc\_harmony \textless{}{-} readRDS("resultados/objects/pbmc\_harmony\
\FunctionTok{saverDS}\NormalTok{(pbmc\_harmony, }\FunctionTok{file.path}\NormalTok{(dir\_objects, }\StringTok{"pbmc\_h"}

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{{}\StringTok{"}\SpecialCharTok{\textbackslash{}\n}\StringTok{[OK] SECTION 3 COMPLETE: M}}
\CommentTok{\# SECTION 4 {-}{-} HARMONY BATCH CORRECTION}
\CommentTok{\# =====}
\CommentTok{\# Harmony adjusts cell embeddings to remove sample{-}level batch effects while}
\CommentTok{\# preserving biological variation. All downstream steps use the "harmony"}
\CommentTok{\# reduction instead of "pca".}

\NormalTok{output\_dir }\OtherTok{\textless{}{-}}\NormalTok{ dir\_01}

\NormalTok{pbmc\_harmony }\OtherTok{\textless{}{-}}\NormalTok{ pbmc\_harmony }\SpecialCharTok{\%\textgreater{}\%}
\FunctionTok{RunHarmony}\NormalTok{{}\StringTok{"orig.ident"}\AttributeTok{plot\_convergence =} \Const
\FunctionTok{RunUMAP}\NormalTok{{}\AttributeTok{reduction =} \StringTok{"harmony"}\AttributeTok{dims =}

\FunctionTok{save\_pdf}\NormalTok{{}\FunctionTok{DimPlot}\NormalTok{(pbmc\_harmony, }\AttributeTok{group.by =} \String
\StringTok{"umap\_postharmony.pdf"}\NormalTok{})}

\CommentTok{\# Checkpoint {-}{-} restore with: pbmc\_harmony \textless{}{-} readRDS("resultados/objects/pbmc\_harmony\
\FunctionTok{saverDS}\NormalTok{(pbmc\_harmony, }\FunctionTok{file.path}\NormalTok{(dir\_objects, }\StringTok{"pbmc\_h"}

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{{}\StringTok{"}\SpecialCharTok{\textbackslash{}\n}\StringTok{[OK] SECTION 4 COMPLETE: H}}
\CommentTok{\# SECTION 5 {-}{-} RESOLUTION OPTIMIZATION}
\CommentTok{\# =====}
\CommentTok{\# Two diagnostics guide the choice of clustering resolution:}
\CommentTok{\# (a) Elbow plot {-}{-} k{-}means within{-}cluster sum of squares across k values.}
\CommentTok{\# The inflection point suggests the number of major cell types.}
\CommentTok{\# (b) Clustree {-}{-} tracks cluster stability across Leiden resolutions.}
\CommentTok{\# Choose the lowest resolution where clusters stop merging.}
\CommentTok{\#}
\CommentTok{\# +{-} PARAMETERS {-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}}
\CommentTok{\# k\_range : k values tested in the elbow plot}
\CommentTok{\# resolutions\_test : Leiden resolutions swept by clustree}
\CommentTok{\# {-}\textgreater{} Inspect elbow\_plot.pdf and clustree.pdf before setting cluster\_resolution}
\CommentTok{\# in Section 6.}
\CommentTok{\# +{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}}
\NormalTok{k\_range }\OtherTok{\textless{}{-}} \DecValTok{1}\SpecialCharTok{:}\DecValTok{31}
\NormalTok{resolutions\_test }\OtherTok{\textless{}{-}} \FunctionTok{c}\NormalTok{{}\FloatTok{0.15}\NormalTok{{}, }\Floa

\NormalTok{output\_dir }\OtherTok{\textless{}{-}}\NormalTok{ dir\_02}

\CommentTok{\# {-}{-} 5a. Elbow plot {-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}}
\NormalTok{pca\_data }\OtherTok{\textless{}{-}} \FunctionTok{Embeddings}\NormalTok{(pbmc\_harmony, }\StringTok{"pca"}\
\NormalTok{wss }\OtherTok{\textless{}{-}} \FunctionTok{apply}\NormalTok{(k\_range, }\ControlFlowTok{function}\No

\NormalTok{elbow\_plot }\OtherTok{\textless{}{-}} \FunctionTok{ggplot}\NormalTok{{}\FunctionTok{data.frame}\NormalTok{{}
\FunctionTok{geom\_line}\NormalTok{{} }\SpecialCharTok{+} \FunctionTok{geom\_point}\NormalTok{{} }\SpecialCharTok{+}
\FunctionTok{labs}\NormalTok{{}\AttributeTok{x =} \StringTok{"Number of clusters (k)"}\AttributeTok{y}
\FunctionTok{theme\_minimal}\NormalTok{{}}
```


[illegible]


```

\FunctionTok{DimPlot}\NormalTok{(pbmc\_harmony, }\AttributeTok{group.by =} \StringTok{"celltype\_curated"}\NormalTok{,
  \AttributeTok{label =} \ConstantTok{TRUE}\NormalTok{, }\AttributeTok{repel =} \ConstantTok{TRUE}\NormalTok{,
  \StringTok{"umap\_curated.pdf"}
\NormalTok{)}

\CommentTok{\# Checkpoint {-}{-} restore with: pbmc\_harmony \textless{-}{-} readRDS("resultados/objects/pbmc\_harmony")
\FunctionTok{saveRDS}\NormalTok{(pbmc\_harmony, }\FunctionTok{file.path}\NormalTok{(dir\_objects, }\StringTok{"pbmc\_h"}

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{({}\StringTok{""}\SpecialCharTok{\textbackslash{n}}\StringTok{[OK] SECTION 12 COMPLETE:}
\CommentTok{\# SECTION 13 {-}{-} EXPORT TO H5AD (Scanpy / Python)}
\CommentTok{\# =====}
\CommentTok{\# Exports the curated object to AnnData h5ad format for Python{-}based}
\CommentTok{\# trajectory and velocity analyses (Scanpy, scFates, Palantir {-}{-} all}
\CommentTok{\# pre{-}installed in the Docker image).}

\FunctionTok{export\_to\_scanpy}\NormalTok{(pbmc\_harmony,}
  \FunctionTok{file.path}\NormalTok{(dir\_objects, }\StringTok{"pbmc\_harmony\_curated.h5ad"}\NormalTok{)}

\CommentTok{\# To export a specific cell type:}
\CommentTok{\# export\_to\_scanpy(}
\CommentTok{\#   subset(pbmc\_harmony, subset = celltype\_curated == "Guard Cell"),}
\CommentTok{\#   file.path(dir\_objects, "GuardCell.h5ad")}
\CommentTok{\# )}

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{({}\StringTok{""}\SpecialCharTok{\textbackslash{n}}\StringTok{[OK] SECTION 13 COMPLETE:}

```

6 capitulo2_pseudobulk_de.R

Script principal de capítulo 2: pseudobulk -> DESeq2 -> volcano -> tablas DE -> GO -> heatmap -> red hdWGCNA.

```
\CommentTok{\# =====}
\CommentTok{\# capitulo2\_pseudobulk\_de.R {-}{-} Pseudobulk Differential Expression \& GO (Part 2)}
\CommentTok{\# =====}
\CommentTok{\# Requires: pbmc\_harmony\_curated.rds generated by capitulo1\_single\_cell.R}
\CommentTok{\# Covers pseudobulk DE (DESeq2), volcano plots, GO enrichment,}
\CommentTok{\# log2FC heatmaps, and network inference.}
\CommentTok{\# =====}


\CommentTok{\# {-}{-} Configuration (must match capitulo1\_single\_cell.R) {-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}{-}}
\NormalTok{PIPELINE\_DIR }OtherTok{textless{}{-}} StringTok{"workspace/workflow/nuevo"}
\NormalTok{DATA\_DIR      }OtherTok{textless{}{-}} StringTok{"workspace/."}
\NormalTok{base\_dir       }OtherTok{textless{}{-}} FunctionTok{file.path}\NormalTok{(DATA\_DIR, )StringTok{"resultado"}}


\FunctionTok{source}\NormalTok{{}\FunctionTok{file.path}\NormalTok{(PIPELINE\_DIR, )StringTok{"load\_libraries.R"}\NormalTok{}}No
\FunctionTok{source}\NormalTok{{}\FunctionTok{file.path}\NormalTok{(PIPELINE\_DIR, )StringTok{"custom\_seurat.R"}\NormalTok{}}Nor
\FunctionTok{source}\NormalTok{{}\FunctionTok{file.path}\NormalTok{(PIPELINE\_DIR, )StringTok{"ScRNA\_Analysis\_Func"}}


\FunctionTok{list2env}\NormalTok{{}\FunctionTok{create\_pipeline\_dirs}\NormalTok{(base\_dir, )AttributeTok{envir =}}
\FunctionTok{set.seed}\NormalTok{{}\DecValTok{1807}\NormalTok{}}


\CommentTok{\# =====}
\CommentTok{\# \#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\#\ PART 2 {-}{-} PSEUDOBULK DE \& GO \#\#\#\#\#\#\#\#\#\#\}
\CommentTok{\# =====}



\CommentTok{\# =====}
\CommentTok{\# LOAD {-}{-} CURATED OBJECT FROM PART 1}
\CommentTok{\# =====}
\CommentTok{\# Loads the final annotated object produced by capitulo1\_single\_cell.R.}
\CommentTok{\# Make sure capitulo1\_single\_cell.R has been run and the checkpoint exists.}




\NormalTok{pbmc\_harmony }OtherTok{textless{}{-}} FunctionTok{readRDS}\NormalTok{{}\FunctionTok{file.path}\NormalTok{}}To


\CommentTok{\# =====}
\CommentTok{\# SECTION 14 {-}{-} CELL{-}TYPE SUBSETS}
\CommentTok{\# =====}
\CommentTok{\# Creates one Seurat subset per curated cell type for downstream pseudobulk}
\CommentTok{\# analysis. Object names are sanitised so they can be used safely as list names}
\CommentTok{\# and output filenames.}
\CommentTok{\#}
\CommentTok{\# Edit here: metadata column with final cell{-}type names.}
\NormalTok{pseudobulk\_annot\_col }OtherTok{textless{}{-}} StringTok{"celltype\_curated"}


\CommentTok{\# Create cell{-}type subsets for pseudobulk analysis}
\NormalTok{cell\_type\_subsets }OtherTok{textless{}{-}} FunctionTok{create\_cell\_type\_subsets}\NormalTok{(pbmc\_h}


\CommentTok{\# =====}


\FunctionTok{message}\NormalTok{{}\StringTok{" }\SpecialCharTok{textbackslash{}\n}\StringTok{[OK] SECTION 14 COMPLETE:"}
\CommentTok{\# SECTION 15 {-}{-} PSEUDO{-}REPLICATE ASSIGNMENT}
\CommentTok{\# =====}
\CommentTok{\# Assigns random pseudo{-}replicates within each condition for every cell type.}
\CommentTok{\# Only subsets containing at least two conditions are kept for Part 2.
```



```

\CommentTok{\# SECTION 19 {-}{-} GO ENRICHMENT (SIMPLE)}
\CommentTok{\# =====}
\CommentTok{\# Gene Ontology enrichment per cell type for the selected contrast.}

\CommentTok{\# Edit here:}
\CommentTok{\#   go\_space = "BP", "MF", or "CC"}
\CommentTok{\#   Arabidopsis: go\_orgdb = org.At.tair.db | go\_keytype = "TAIR"}
\NormalTok{go\_space      }\OtherTok{\textless{}\{-}\} \StringTok{"BP"}
\NormalTok{go\_orgdb        }\OtherTok{\textless{}\{-}\} \NormalTok{org.At.tair.db}
\NormalTok{go\_keytype      }\OtherTok{\textless{}\{-}\} \StringTok{"TAIR"}
\NormalTok{padj\_cutoff     }\OtherTok{\textless{}\{-}\} \FloatTok{0.05}

\NormalTok{go\_results }\OtherTok{\textless{}\{-}\} \FunctionTok{run\_go\_enrichment\_for\_contrast}\NormalTok{({}
  \AttributeTok{results\_dir  =} \FunctionTok{file.path}\NormalTok{(dir\_06, volcano\_tag),}
  \AttributeTok{output\_dir    =} \FunctionTok{file.path}\NormalTok{(dir\_07, volcano\_tag),}
  \AttributeTok{orgdb          =} \NormalTok{go\_orgdb,}
  \AttributeTok{keytype        =} \NormalTok{go\_keytype,}
  \AttributeTok{go\_space      =} \NormalTok{go\_space,}
  \AttributeTok{padj\_cutoff    =} \NormalTok{padj\_cutoff,}
  \AttributeTok{contrast\_tag  =} \NormalTok{volcano\_tag}
\NormalTok{)})

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{({}\StringTok{"}\SpecialCharTok{\textbackslash{}n}\StringTok{[OK] SECTION 19 COMPLETE:}
\CommentTok{\# SECTION 20 {-}{-} LOG2FC HEATMAP}
\CommentTok{\# =====}
\CommentTok{\# Heatmap of log2FC values across all cell types for the selected contrast.}
\CommentTok{\# Genes are ordered by the internal hierarchical dendrogram (cluster\_rows = TRUE).}
\CommentTok{\#}
\CommentTok{\# Edit here: color scale range for log2FC values.}
\NormalTok{heatmap\_limits }\OtherTok{\textless{}\{-}\} \FunctionTok{c}\NormalTok{({}\SpecialCharTok{\{-}\}\DecValTok{5}\NormalTok{)}

\FunctionTok{build\_logfc\_heatmap}\NormalTok{({}
  \AttributeTok{logfc\_table  =} \NormalTok{diff\_tables}\SpecialCharTok{$}\NormalTok{logfc,}
  \AttributeTok{contrast\_tag  =} \NormalTok{volcano\_tag,}
  \AttributeTok{output\_dir    =} \FunctionTok{file.path}\NormalTok{(dir\_06, volcano\_tag),}
  \AttributeTok{limits         =} \NormalTok{heatmap\_limits}
\NormalTok{)})

\CommentTok{\# =====}

\FunctionTok{message}\NormalTok{({}\StringTok{"}\SpecialCharTok{\textbackslash{}n}\StringTok{[OK] SECTION 20 COMPLETE:}
\CommentTok{\# SECTION 21 {-}{-} COEXPRESSION NETWORK (hdWGCNA) ON SIGNIFICANT GENES}
\CommentTok{\# =====}
\CommentTok{\# Builds one hdWGCNA network using the complete significant{-}gene subset from the}
\CommentTok{\# log2FC heatmap table created in Section 20.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\#   n\_metacells = metacells per celltype x sample}
\CommentTok{\#   soft\_power  = NULL for automatic detection, or use a number}
\NormalTok{wgcn\_name }\OtherTok{\textless{}\{-}\} \StringTok{"unified"}
\NormalTok{n\_metacells }\OtherTok{\textless{}\{-}\} \DecValTok{25}
\NormalTok{soft\_power  }\OtherTok{\textless{}\{-}\} \ConstantTok{NULL}

\FunctionTok{run\_unified\_hdwgcna}\NormalTok{({}
  \AttributeTok{seurat\_obj    =} \NormalTok{pbmc\_harmony,}
  \AttributeTok{de\_table\_path =} \FunctionTok{file.path}\NormalTok{(dir\_06, volcano\_tag, }\StringTok{"tabla\_log2FC"}
  \AttributeTok{output\_dir      =} \NormalTok{dir\_08,}
  \AttributeTok{annot\_col       =} \NormalTok{pseudobulk\_annot\_col,}

```

```

\AttributeTok{sample\_col      =} \StringTok{"orig.ident"}\NormalTok{,{,}
\AttributeTok{wgcn\_name      =}\NormalTokTok{ wgcn\_name,{,}
\AttributeTok{n\_metacells    =}\NormalTokTok{ n\_metacells,{,}
\AttributeTok{soft\_power     =}\NormalTokTok{ soft\_power}
\NormalTokTok{)}

\FunctionTok{message}\NormalTokTok{({}\StringTok{""}\SpecialCharTok{\textbackslash{n}}\StringTok{\textbackslash{u2713} SECT

\CommentTok{\# =====}
\CommentTok{\# SECTION 22 {-}{-} NETWORK EXPORT \& VISUALIZATION}
\CommentTok{\# =====}
\CommentTok{\# Reads the unified hdWGCNA object from Section 21 and saves:}
\CommentTok{\#   edges, nodes and a co{-}expression network PDF.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\#   tom\_threshold = lower gives more edges; higher gives simpler networks}
\CommentTok{\#   n\_hub\_label = number of hub genes labelled in the plot}
\NormalTokTok{tom\_threshold }\OtherTok{\textless{}{-}} \FloatTok{0.5}
\NormalTokTok{n\_hub\_label   }\OtherTok{\textless{}{-}} \DecValTok{5}

\FunctionTok{plot\_hdwgcna\_network}\NormalTokTok{({}
  \AttributeTok{hdwgcna\_dir    =}\NormalTokTok{ dir\_08,{,}
  \AttributeTok{output\_dir      =}\NormalTokTok{ dir\_08,{,}
  \AttributeTok{tom\_threshold  =}\NormalTokTok{ tom\_threshold,{,}
  \AttributeTok{n\_hub\_label    =}\NormalTokTok{ n\_hub\_label}
\NormalTokTok{)}

\FunctionTok{message}\NormalTokTok{({}\StringTok{""}\SpecialCharTok{\textbackslash{n}}\StringTok{[OK] SECTION 22 COMPLETE:}

```

7 load_libraries_python.py

Carga de paquetes Python (capítulo 3: scanpy, scFates, Palantir).

```
\CommentTok{\# =====}
\CommentTok{\# load\_libraries\_python.py {-}{-} Python dependencies for pseudotime analysis}
\CommentTok{\# =====}
\CommentTok{\# Loaded automatically by capitulo3\_pseudotime.ipynb.}
\CommentTok{\# Do not modify unless adding new packages.}

\ImportTok{import}\NormalTok{ os}
\ImportTok{import}\NormalTok{ sys}
\ImportTok{import}\NormalTok{ warnings}

\ImportTok{import}\NormalTok{ numpy }\ImportTok{as}\NormalTok{ np}
\ImportTok{import}\NormalTok{ pandas }\ImportTok{as}\NormalTok{ pd}
\ImportTok{from}\NormalTok{ scipy.stats }\ImportTok{import}\NormalTok{ pearsonr}
\ImportTok{from}\NormalTok{ sklearn.preprocessing }\ImportTok{import}\NormalTok{ scale}

\ImportTok{import}\NormalTok{ scanpy }\ImportTok{as}\NormalTok{ sc}
\ImportTok{import}\NormalTok{ scFates }\ImportTok{as}\NormalTok{ scf}
\ImportTok{import}\NormalTok{ palantir}
\ImportTok{from}\NormalTok{ IPython }\ImportTok{import}\NormalTok{ get\_ipython}

\ImportTok{import}\NormalTok{ matplotlib}

\NormalTok{shell }\OperatorTok{=}\NormalTok{ get\_ipython()}
\ControlFlowTok{if}\NormalTok{ shell }\KeywordTok{is} \KeywordTok{not} \VariableTok{None} \KeywordTok{and} \StringTok{"matplotlib"}\NormalTok{ }\StringTok{"inline"}\NormalTok{}}
\ControlFlowTok{else}\NormalTok{:{}}
\NormalTok{ matplotlib.use()\StringTok{"Agg"}\NormalTok{ } }\CommentTok{\# non{-}interactive backend for script exe}

\ImportTok{import}\NormalTok{ matplotlib.pyplot }\ImportTok{as}\NormalTok{ plt}
\ImportTok{import}\NormalTok{ matplotlib.patches }\ImportTok{as}\NormalTok{ mpatches}
\ImportTok{import}\NormalTok{ seaborn }\ImportTok{as}\NormalTok{ sns}

\NormalTok{warnings.filterwarnings()\StringTok{"ignore"}\NormalTok{}}
\NormalTok{sc.settings.verbosity }\OperatorTok{=}\NormalTok{ DecValTok{3}}
\NormalTok{sc.settings.logfile }\OperatorTok{=}\NormalTok{ sys.stdout}
```

8 ScrNA_Pseudotime_Functions.py

Funciones núcleo de capítulo 3: carga de objeto curado, selección de cell types, inferencia de trayectoria (PPT/ElPiGraph), gene trends.

```
\CommentTok{\# =====}
\CommentTok{\# ScrNA\_Pseudotime\_Functions.py {-}{-} simple helpers for Capitulo 3}
\CommentTok{\# =====}
\CommentTok{\# Companion to capitulo3\_pseudotime.ipynb.}
\CommentTok{\# This file intentionally contains only the functions used by the notebook:}
\CommentTok{\# STEP 25 {-}{-} load data}
\CommentTok{\# STEP 26 {-}{-} choose cell types}
\CommentTok{\# STEP 27 {-}{-} build pseudotime trajectory}
\CommentTok{\# STEP 29 {-}{-} top{-}N and custom{-}highlight gene trends}
\CommentTok{\# =====}

\CommentTok{\# STEP 25 {-}{-} load\_curated\_object}
\CommentTok{\# =====}
\NormalTok{PLOT\_FIGSIZE }\OperatorTok{=}\NormalTok{( }\DecValTok{18}\NormalTok{, }\DecValTok{18}\NormalTok{)}
\NormalTok{PLOT\_DPI }\OperatorTok{=}\NormalTok{ }\DecValTok{200}

\KeywordTok{def}\NormalTok{ save\_plot\_18x18(fig, path):}
\NormalTok{    fig.set\_size\_inches(}\OperatorTok{*}\NormalTok{PLOT\_FIGSIZE, forward)\OperatorTok{=}\VariableTok{True}}
\NormalTok{    fig.savefig(path, dpi}\OperatorTok{=}\NormalTok{PLOT\_DPI, facecolor}\OperatorTok{=}\StringTok{"white"}
        \ControlFlowTok{return}\NormalTok{ path}

\CommentTok{\# Notebook wrapper for Step 25. Loads the full curated object, fixes Seurat{-}style}
\CommentTok{\# coordinate names, plots the overview UMAP, and prints available cell types.}
\KeywordTok{def}\NormalTok{ load\_curated\_object(input\_h5ad, dir\_pseudotime, annotation\_col, n\_jobs)\OperatorTok{=}\NormalTok{ }

    \BuiltInTok{print}\NormalTok{(}\SpecialStringTok{f"Loading object: "}\NormalTok{input\_h5ad}\NormalTok{)}
\NormalTok{    adata }\OperatorTok{=}\NormalTok{sc.read\_h5ad(input\_h5ad)}
\CommentTok{\# Convert obsm keys from R/Seurat format to scanpy format}
\CommentTok{\# Handles both DataFrame (from older anndata) and ndarray (from newer anndata)}
\ControlFlowTok{for}\NormalTok{ key\_from, key\_to }\KeywordTok{in}\NormalTok{ [}\StringTok{"UMAP"}\NormalTok{, }
    \ControlFlowTok{if}\NormalTok{ key\_from }\KeywordTok{in}\NormalTok{ adata.obsm }\KeywordTok{and}\NormalTok{ key\_to }\KeywordTok{in}\NormalTok{ adata.obsm:}
\NormalTok{    val }\OperatorTok{=}\NormalTok{ adata.obsm[key\_from]}
\NormalTok{    adata.obsm[key\_to] }\OperatorTok{=}\NormalTok{ val.values }\ControlFlowTok{if}\NormalTok{ key\_to }\BuiltInTok{hasattr}\NormalTok{(adata.obsm[key\_to], "dtype")}

\NormalTok{    sc.settings.figdir }\OperatorTok{=}\NormalTok{ dir\_pseudotime}
\NormalTok{    sc.set\_figure\_params(figsize}\OperatorTok{=}\NormalTok{PLOT\_FIGSIZE, dpi}\OperatorTok{=}\DecValTok{200}

\NormalTok{    fig }\OperatorTok{=}\NormalTok{sc.pl.umap(}
\NormalTok{        adata,}
\NormalTok{        color }\OperatorTok{=}\NormalTok{annotation\_col,}
\NormalTok{        legend\_loc }\OperatorTok{=}\StringTok{"on data"}\NormalTok{,}
\NormalTok{        legend\_fontsize }\OperatorTok{=}\DecValTok{6}\NormalTok{,}
\NormalTok{        legend\_fontoutline }\OperatorTok{=}\DecValTok{2}\NormalTok{,}
\NormalTok{        frameon }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{        size }\OperatorTok{=}\DecValTok{10}\NormalTok{,}
\NormalTok{        show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{        return\_fig }\OperatorTok{=}\VariableTok{True}\NormalTok{,}
\NormalTok{    )}

    \CommentTok{\# Add a side legend with one color swatch per cell type, in addition to}
    \CommentTok{\# the on{-}data labels above (scanpy only supports one legend\_loc at a time).}
\NormalTok{    ax }\OperatorTok{=}\NormalTok{fig.axes[}\DecValTok{0}\NormalTok{]}
\NormalTok{    cats }\OperatorTok{=}\NormalTok{adata.obs[annotation\_col].cat.categories}
\NormalTok{    colors }\OperatorTok{=}\NormalTok{adata.uns[}\SpecialStringTok{f"}\NormalTok{annotation\_col}\NormalTok{]}
\NormalTok{    handles }\OperatorTok{=}\NormalTok{ax.legend(handles=}
```

```

\NormalTok{      plt.Line2D([\DecValTok{0}\NormalTok{[], [\DecValTok{0}\NormalTok{[], marker}\OperatorTok{=}\StringTok{
  \ControlFlowTok{for}\NormalTok{ c }\KeywordTok{in}\NormalTok{ colors}
\NormalTok{    ]}
\NormalTok{  ax.legend()
\NormalTok{    handles, cats,}
\NormalTok{    loc}\OperatorTok{=}\StringTok{"center left"}\NormalTok{, bbox\_to\_anchor}\OperatorTok{=}\NormalTok{
\NormalTok{    fontsize}\OperatorTok{=}\DecValTok{6}\NormalTok{, frameon}\OperatorTok{=}\VariableTok{False}\NormalTok{
\NormalTok{  })}

\NormalTok{  fig.subplots\_adjust(right)\OperatorTok{=}\FloatTok{0.78}\NormalTok{}}
\NormalTok{  save\_plot\_18x18(fig, os.path.join(dir\_pseudotime, }\StringTok{"umap\_overview.png"}\NormalTok{))}
\NormalTok{  save\_plot\_18x18(fig, os.path.join(dir\_pseudotime, }\StringTok{"umap\_overview.pdf"}\NormalTok{))}
\NormalTok{  plt.close(fig)}

\NormalTok{  counts }\OperatorTok{=}\NormalTok{adata.obs[annotation\_col].value\_counts()}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"}\CharTok{\textbackslash{n}}\SpecialStringTok{Cell types in te
\ControlFlowTok{for}\NormalTok{ i, ct }\KeywordTok{in} \BuiltInTok{enumerate}\NormalTok{({}\BuiltInTok{sorted}\NormalTok{
  \BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"  }\SpecialCharTok{\{}\}\NormalTok{{i}\SpecialCharTok{:2d}\}\S

\BuiltInTok{print}\NormalTok{({}\StringTok{"}\CharTok{\textbackslash{n}}\StringTok{STEP 25 COMPLETE: full object lo
\ControlFlowTok{return}\NormalTok{adata, n\_jobs}

\CommentTok{\# =====}
\CommentTok{\# STEP 26 {-}{-} preview\_trajectory\_selection}
\CommentTok{\# =====}
\CommentTok{\# Notebook wrapper for Step 26. Subsets the selected cell types and displays a}
\CommentTok{\# UMAP preview so the user can decide whether the selection makes biological sense.}
\KeywordTok{def}\NormalTok{ preview\_trajectory\_selection(adata, clusters, annotation\_col, dir\_pseudotime):}
\NormalTok{  missing }\OperatorTok{=}\NormalTok{[ x }\ControlFlowTok{for}\NormalTok{ x }\KeywordTok{in}\NormalTok{ c
  \ControlFlowTok{if}\NormalTok{ missing:}
    \ControlFlowTok{raise} \PreprocessorTok{ValueError}\NormalTok{({}
      \StringTok{"These cell types were not found in "}
      \SpecialStringTok{f"\textquotesingle{}\}\SpecialCharTok{\{}\}\NormalTok{annotation\_col}\SpecialCharTok{\}\}\S
\NormalTok{  })}

\NormalTok{  adata\_sub }\OperatorTok{=}\NormalTok{adata[adata.obs[annotation\_col].isin(clusters)].copy()}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"Selected }\SpecialCharTok{\{}\}\BuiltInTok{len}\NormalTok{({}adata

\NormalTok{  fig }\OperatorTok{=}\NormalTok{sc.pl.umap(}
\NormalTok{    adata\_sub,}
\NormalTok{    color }\OperatorTok{=}\NormalTok{annotation\_col,}
\NormalTok{    legend\_loc }\OperatorTok{=}\StringTok{"on data"}\NormalTok{,}
\NormalTok{    legend\_fontsize }\OperatorTok{=}\DecValTok{6}\NormalTok{,}
\NormalTok{    legend\_fontoutline }\OperatorTok{=}\DecValTok{2}\NormalTok{,}
\NormalTok{    frameon }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{    size }\OperatorTok{=}\DecValTok{10}\NormalTok{,}
\NormalTok{    show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{    return\_fig }\OperatorTok{=}\VariableTok{True}\NormalTok{,}
\NormalTok{  })}

\CommentTok{\# Side legend with one color swatch per cell type, in addition to the}
\CommentTok{\# on{-}data labels above (scanpy only supports one legend\_loc at a time).}
\NormalTok{  ax }\OperatorTok{=}\NormalTok{fig.axes[\DecValTok{0}\NormalTok{]}
\NormalTok{  cats }\OperatorTok{=}\NormalTok{adata\_sub.obs[annotation\_col].cat.categories}
\NormalTok{  colors }\OperatorTok{=}\NormalTok{adata\_sub.uns[\SpecialStringTok{f"}\SpecialCharTok{\{}\}\NormalTok{a
\NormalTok{  handles }\OperatorTok{=}\NormalTok{[}
\NormalTok{    plt.Line2D([\DecValTok{0}\NormalTok{[], [\DecValTok{0}\NormalTok{[], marker}\OperatorTok{=}\StringTok{
  \ControlFlowTok{for}\NormalTok{ c }\KeywordTok{in}\NormalTok{ colors}
\NormalTok{  ]}
\NormalTok{  ax.legend()

```

```

\NormalTok{      handles, cats,}
\NormalTok{      loc}\OperatorTok{=}\StringTok{"center left"}\NormalTok{, bbox\_to\_anchor}\OperatorTok{=}\NormalTok{
\NormalTok{      fontsize}\OperatorTok{=}\DecValTok{6}\NormalTok{, frameon}\OperatorTok{=}\VariableTok{False}\NormalTok{
\NormalTok{    })}

\NormalTok{    fig.subplots\_adjust(right)\OperatorTok{=}\FloatTok{0.78}\NormalTok{}}
\NormalTok{    save\_plot\_18x18(fig, os.path.join(dir\_pseudotime, }\StringTok{"umap\_selection.png"}\NormalTok{))}
\NormalTok{    save\_plot\_18x18(fig, os.path.join(dir\_pseudotime, }\StringTok{"umap\_selection.pdf"}\NormalTok{))}
\NormalTok{    plt.close(fig)}

\BuiltInTok{print}\NormalTok{({}\StringTok{""}\CharTok{\textbackslash}\StringTok{STEP 26 COMPLETE: continue only}
\ControlFlowTok{return}\NormalTok{adata\_sub}

\CommentTok{\# =====}
\CommentTok{\# STEP 27 {-}{-} trajectory\_run}
\CommentTok{\# =====}
\CommentTok{\# Small helper to describe one trajectory parameter set as a dict. Used to}
\CommentTok{\# build the TRAJECTORY\_RUNS list in the notebook\textquotesingle{s Step 27 cell.}}
\CommentTok{\# The run\textquotesingle{s name/folder is generated from the parameters themselves (not a}
\CommentTok{\# free\textquotesingle{-}text label), so two runs are named the same only if every parameter}
\CommentTok{\# matches {-}{-} in that case they are the same run and should share a folder.}}
\CommentTok{\# seed controls the stochastic steps inside build\_pseudotime\_trajectory}
\CommentTok{\# (neighbors, ForceAtlas2 layout, tree fitting) {-}{-} same seed + same params}
\CommentTok{\# always reproduces the same result.}}
\KeywordTok{def}\NormalTok{ trajectory\_run(nodes)\OperatorTok{=}\DecValTok{50}\NormalTok{, sigma}\OperatorTok{=}\Floa
\NormalTok{    name }\OperatorTok{=}\StringTok{"f"}\NormalTok{nodes}\SpecialCharTok{\}\NormalTok{nodes}\SpecialCharTok{\}}\NormalTok{
\ControlFlowTok{return}\NormalTok{ {}}
\StringTok{"name"}\NormalTok{: name,}
\StringTok{"nodes"}\NormalTok{: nodes,}
\StringTok{"sigma"}\NormalTok{: sigma,}
\StringTok{"lambda"}\NormalTok{: lambda\_value,}
\StringTok{"eigs"}\NormalTok{: eigs,}
\StringTok{"seed"}\NormalTok{: seed,}
\NormalTok{ }}

\CommentTok{\# =====}
\CommentTok{\# STEP 27 {-}{-} build\_pseudotime\_trajectory}
\CommentTok{\# =====}
\CommentTok{\# Full pipeline: subset cells {-}\textgreater{} Palantir diffusion maps {-}\textgreater{} scFates tree {-}
\CommentTok{\# root selection {-}\textgreater{} pseudotime assignment. Called by run\_trajectory\_runs,}
\CommentTok{\# which is what the notebook\textquotesingle{s Step 27 cell actually calls.}}
\KeywordTok{def}\NormalTok{ build\_pseudotime\_trajectory()}
\NormalTok{    adata,}
\NormalTok{    clusters, }\CommentTok{\# list of cell types to include (values from annotation\_col)}
\NormalTok{    root\_cluster, }\CommentTok{\# cell type where pseudotime = 0 (the biological progenitor)}
\NormalTok{    annotation\_col, }\CommentTok{\# adata.obs column containing cell type labels}
\NormalTok{    nodes }\OperatorTok{=}\DecValTok{150}\NormalTok{, }\CommentTok{\# tree nodes: more = finer bran}
\NormalTok{    sigma }\OperatorTok{=}\FloatTok{0.2}\NormalTok{, }\CommentTok{\# smoothing: lower = tree follow}
\NormalTok{    ppt\_lambda }\OperatorTok{=}\DecValTok{60}\NormalTok{, }\CommentTok{\# complexity: higher = simpler}
\NormalTok{    n\_components}\OperatorTok{=}\DecValTok{50}\NormalTok{, }\CommentTok{\# total diffusion map dimension}
\NormalTok{    n\_eigs }\OperatorTok{=}\DecValTok{20}\NormalTok{, }\CommentTok{\# dimensions retained (must be}
\NormalTok{    n\_neighbors }\OperatorTok{=}\DecValTok{50}\NormalTok{, }\CommentTok{\# neighbors for final graph (h}
\NormalTok{    seed }\OperatorTok{=}\DecValTok{3}\NormalTok{, }\CommentTok{\# random seed for reproducibili}
\NormalTok{():}
\CommentTok{\# Subset to selected cell types}
\ControlFlowTok{if}\NormalTok{ clusters:}
\NormalTok{    adata }\OperatorTok{=}\NormalTok{ adata[adata.obs[annotation\_col].isin(clusters)].copy()}

\CommentTok{\# Palantir diffusion maps: captures global developmental geometry}

```



```

\NormalTok{      pca\_proj } \OperatorTok{=} \NormalTok{ pd.DataFrame(adata.obsm[\StringTok{"X\_pca"}] \NormalTok{, index}
\NormalTok{dm\_res } \OperatorTok{=} \NormalTok{ palantir.utils.run\_diffusion\_maps(pca\_proj, n\_components) \OperatorTok{=}
\NormalTok{ms\_data } \OperatorTok{=} \NormalTok{ palantir.utils.determine\_multiscale\_space(}
\NormalTok{      dm\_res, n\_eigs) \OperatorTok{=} \BuiltInTok{min} \NormalTok{(n\_eigs, n\_components ) \OperatorTok{=}
\NormalTok{    )}
\NormalTok{val } \OperatorTok{=} \NormalTok{ ms\_data.values } \ControlFlowTok{if} \BuiltInTok{hasattr} \NormalTok{(ms\_data.obsm[\StringTok{"X\_palantir"}] \OperatorTok{=} \NormalTok{ val}

\CommentTok{ \# Graph in Palantir space for tree construction}
\NormalTok{sc.pp.neighbors(adata, n\_neighbors) \OperatorTok{=} \NormalTok{(n\_neighbors, use\_rep) \OperatorTok{=} \NormalTok{St
\NormalTok{adata.obsm[\StringTok{"X\_pca2d"}] \NormalTok{ } \OperatorTok{=} \NormalTok{ adata.obsm[\StringTok{"X\_p
\NormalTok{sc.tl.draw\_graph(adata, init\_pos) \OperatorTok{=} \StringTok{"X\_pca2d"} \NormalTok{}}

\CommentTok{ \# Build principal tree (PPT = Principal Polynomial Tree)}
\NormalTok{scf.tl.tree(}
\NormalTok{    adata,}
\NormalTok{    method } \OperatorTok{=} \StringTok{"ppt"} \NormalTok{,}
\NormalTok{    Nodes } \OperatorTok{=} \NormalTok{ nodes,}
\NormalTok{    use\_rep } \OperatorTok{=} \StringTok{"palantir"} \NormalTok{,}
\NormalTok{    plot } \OperatorTok{=} \VariableTok{False} \NormalTok{,}
\NormalTok{    device } \OperatorTok{=} \StringTok{"cpu"} \NormalTok{,}
\NormalTok{    seed } \OperatorTok{=} \NormalTok{ seed,}
\NormalTok{    ppt\_lambda } \OperatorTok{=} \NormalTok{ ppt\_lambda,}
\NormalTok{    ppt\_nsteps } \OperatorTok{=} \DecValTok{200} \NormalTok{,}
\NormalTok{    ppt\_sigma } \OperatorTok{=} \NormalTok{ sigma,}
\NormalTok{  )}

\CommentTok{ \# Root selection: most{-}connected cell within root\_cluster}
\NormalTok{mask } \OperatorTok{=} \NormalTok{ (adata.obs[annotation\_col] ) \OperatorTok{=} \NormalTok{ root
\NormalTok{cluster\_cells } \OperatorTok{=} \NormalTok{ adata.obs\_names[mask]}
\NormalTok{sub\_conn } \OperatorTok{=} \NormalTok{ adata.obsp[\StringTok{"connectivities"}] [mask] [:
\NormalTok{degrees } \OperatorTok{=} \NormalTok{ np.array(sub\_conn.) \BuiltInTok{sum} \NormalTok{(axis) \OperatorTok{=}
\NormalTok{root\_cell } \OperatorTok{=} \NormalTok{ cluster\_cells[degrees.argmax()] }

\NormalTok{adata.obs[\StringTok{"is\_root"}] \NormalTok{ } \OperatorTok{=} \NormalTok{ adata.obs\_names } \OperatorTok{=}
\NormalTok{scf.tl.root(adata, root) \OperatorTok{=} \StringTok{"is\_root"} \NormalTok{}}
\NormalTok{scf.tl.pseudotime(adata)}

\BuiltInTok{print} \NormalTok{(\StringTok{"[OK] Trajectory built {-}{-} root cell: "} \SpecialCharTok{\} \NormalTok{)
\ControlFlowTok{return} \NormalTok{ adata}

\CommentTok{ \# =====}
\CommentTok{ \# STEP 27 {-}{-} output helpers for run\_trajectory\_runs}
\CommentTok{ \# =====}
\CommentTok{ \# Adds a dendrogram to an adata object that already has a scFates tree.}
\KeywordTok{def} \NormalTok{ build\_dendrogram(adata):}
\NormalTok{    scf.tl.dendrogram(adata)}
\ControlFlowTok{return} \NormalTok{ adata}

\CommentTok{ \# Saves the force{-}directed tree graph colored by cell type.}
\KeywordTok{def} \NormalTok{ plot\_trajectory\_graphs(}
\NormalTok{    adata,}
\NormalTok{    name,}
\NormalTok{    output\_dir,}
\NormalTok{    annotation\_col,}
\NormalTok{    show\_inline} \OperatorTok{=} \VariableTok{False} \NormalTok{,}
\NormalTok{    param\_label} \OperatorTok{=} \VariableTok{None} \NormalTok{,}
\NormalTok{  ):}
\NormalTok{    os.makedirs(output\_dir, exist\_ok) \OperatorTok{=} \VariableTok{True} \NormalTok{}}

```

```

\NormalTok{      title\_suffix }\OperatorTok{=} \SpecialStringTok{f"}\CharTok{\textbackslash{n}}\SpecialCharTok{\{\}\NormalTok{

\NormalTok{      fig, ax }\OperatorTok{=}\NormalTok{ plt.subplots(figsize)\OperatorTok{=}\NormalTok{PLOT\_FIGSIZE)}
\NormalTok{      scf.pl.graph(adata, color\_cells)\OperatorTok{=}\NormalTok{annotation\_col, ax}\OperatorTok{=}\NormalTok{
\NormalTok{      ax.set\_title}\SpecialStringTok{f"}\SpecialCharTok{\{\}\NormalTok{ax}\SpecialCharTok{.}\NormalTok{get\_
\ControlFlowTok{for}\NormalTok{ ext }\KeywordTok{in}\NormalTok{ [}\StringTok{"pdf"}\NormalTok{, }\StringTok{"png"}
\NormalTok{      fig.savefig(os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\{\}\NormalTok{name}\Spe

\NormalTok{      plt.close(fig)}

\BuiltInTok{print}\NormalTok{(\}\SpecialStringTok{f"[OK] Trajectory graph saved to }\SpecialCharTok{\{\}\NormalTok{o

\KeywordTok{def}\NormalTok{ export\_pseudotime\_table(adata, name, output\_dir, annotation\_col)\OperatorTok{=}\Variab
\NormalTok{      os.makedirs(output\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{()}}
\NormalTok{      cell\_time }\OperatorTok{=}\NormalTok{ adata.obs[}\StringTok{"t"}\NormalTok{[]].copy()}}
\NormalTok{      cell\_time.index.name }\OperatorTok{=}\StringTok{"cell\_id"}
\ControlFlowTok{if}\NormalTok{ annotation\_col }\KeywordTok{is}\KeywordTok{not}\VariableTok{None}\KeywordTok{and}
\NormalTok{      cell\_time[annotation\_col] }\OperatorTok{=}\NormalTok{ adata.obs[annotation\_col]}
\NormalTok{      out\_file }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\
\NormalTok{      cell\_time.to\_csv(out\_file, sep)\OperatorTok{=}\StringTok{""}\CharTok{\textbackslash{t}}\StringTok{""}\
\BuiltInTok{print}\NormalTok{(\}\SpecialStringTok{f"Pseudotime table saved: }\SpecialCharTok{\{\}\NormalTok{out\_fil
\ControlFlowTok{return}\NormalTok{ out\_file}

\CommentTok{## Overlay the principal graph and pseudotime trajectory on the FA layout.}
\KeywordTok{def}\NormalTok{ plot\_pseudotime\_trajectory}
\NormalTok{      adata,}
\NormalTok{      name,}
\NormalTok{      output\_dir,}
\NormalTok{      annotation\_col,}
\NormalTok{      show\_inline}\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{      param\_label}\OperatorTok{=}\VariableTok{None}\NormalTok{,}
\NormalTok{      ):}
\NormalTok{      os.makedirs(output\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{()}}

\NormalTok{      fig, ax }\OperatorTok{=}\NormalTok{ plt.subplots(figsize)\OperatorTok{=}\NormalTok{PLOT\_FIGSIZE)}
\NormalTok{      scf.pl.graph(}
\NormalTok{          adata,}
\NormalTok{          basis }\OperatorTok{=}\StringTok{"draw\_graph\_fa"}\NormalTok{,}
\NormalTok{          color\_cells }\OperatorTok{=}\NormalTok{ annotation\_col,}
\NormalTok{          ax }\OperatorTok{=}\NormalTok{ ax,}
\NormalTok{          show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{      )}
\NormalTok{      scf.pl.trajectory(}
\NormalTok{          adata,}
\NormalTok{          color\_seg }\OperatorTok{=}\StringTok{"t"}\NormalTok{,}
\NormalTok{          basis }\OperatorTok{=}\StringTok{"draw\_graph\_fa"}\NormalTok{,}
\NormalTok{          frameon }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{          s }\OperatorTok{=}\DecValTok{50}\NormalTok{,}
\NormalTok{          scale\_path }\OperatorTok{=}\FloatTok{0.6}\NormalTok{,}
\NormalTok{          ax }\OperatorTok{=}\NormalTok{ ax,}
\NormalTok{          show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{      )}
\NormalTok{      title }\OperatorTok{=}\SpecialStringTok{f"}\SpecialCharTok{\{\}\NormalTok{name}\SpecialCharTok{\{\}\Spec
\ControlFlowTok{if}\NormalTok{ param\_label:}
\NormalTok{      title }\OperatorTok{+=}\SpecialStringTok{f"}\CharTok{\textbackslash{n}}\SpecialCharTok{\{\}\NormalTok{
\NormalTok{      ax.set\_title(title)}
\NormalTok{      save\_plot\_18x18(fig, os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\{\}\NormalTok{na
\NormalTok{      save\_plot\_18x18(fig, os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\{\}\NormalTok{na

```

```

\ControlFlowTok{if}\NormalTok{ show\_inline:}
\ControlFlowTok{pass}
\NormalTok{ plt.close(fig)}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"Pseudotime trajectory saved to "}\SpecialCharTok{\{}\}\NormalTok{out\_file})}

\CommentTok{\# QC plot: show the automatically selected root cell as a red point on FA1/FA2.}
\KeywordTok{def}\NormalTok{ plot\_root\_cell(adata, name, output\_dir, show\_inline)\OperatorTok{=}\VariableTok{False}}
\NormalTok{ os.makedirs(output\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{({})}
\ControlFlowTok{if} \StringTok{"is\_root"} \KeywordTok{not} \KeywordTok{in}\NormalTok{ adata.obs.columns:}
\BuiltInTok{print}\NormalTok{({}\StringTok{"Root{-}cell plot skipped: is\_root not found."}\NormalTok{({})})}
\ControlFlowTok{return} \VariableTok{None}

\NormalTok{ root\_cells }\OperatorTok{=}\NormalTok{ adata.obs\_names[adata.obs[}\StringTok{"is\_root"}]\NormalTok{[0]}
\ControlFlowTok{if} \BuiltInTok{len}\NormalTok{ (root\_cells) }\OperatorTok{==} \DecValTok{0}\NormalTok{(:}
\BuiltInTok{print}\NormalTok{({}\StringTok{"Root{-}cell plot skipped: no root cell marked."}\NormalTok{({})})}
\ControlFlowTok{return} \VariableTok{None}

\NormalTok{ root\_cell }\OperatorTok{=}\NormalTok{ root\_cells[}\DecValTok{0}\NormalTok{]}
\NormalTok{ plot\_data }\OperatorTok{=}\NormalTok{ adata.copy()}
\NormalTok{ plot\_data.obs[}\StringTok{"root\_cell"}]\NormalTok{[0]} \OperatorTok{=}\NormalTok{ np.where(plot\_data.obs[}\StringTok{"root\_cell"}]\NormalTok{[0] == 0)[0]}
\NormalTok{ plot\_data }\OperatorTok{=}\NormalTok{ plot\_data[plot\_data.obs.sort\_values[}\StringTok{"root\_cell"}]\NormalTok{[0]}
\NormalTok{ plot\_data.uns[}\StringTok{"root\_cell\_colors"}]\NormalTok{[0]} \OperatorTok{=}\NormalTok{ plot\_data.uns[}\StringTok{"root\_cell\_colors"}]\NormalTok{[0]}

\NormalTok{ title }\OperatorTok{=}\SpecialStringTok{f"Root cell: "}\SpecialCharTok{\{}\}\NormalTok{ root\_cell }\SpecialStringTok{"}
\ControlFlowTok{if}\NormalTok{ param\_label:}
\NormalTok{ title }\OperatorTok{+=}\SpecialStringTok{f"}\CharTok{\textbackslash n}\SpecialCharTok{\{}\}\NormalTok{ param\_label }\SpecialStringTok{"}

\NormalTok{ fig }\OperatorTok{=}\NormalTok{ sc.pl.embedding(}
\NormalTok{ plot\_data,}
\NormalTok{ basis }\OperatorTok{=}\StringTok{"draw\_graph\_fa"}\NormalTok{,}
\NormalTok{ color }\OperatorTok{=}\StringTok{"root\_cell"}\NormalTok{,}
\NormalTok{ legend\_loc }\OperatorTok{=}\StringTok{"right margin"}\NormalTok{,}
\NormalTok{ title }\OperatorTok{=}\NormalTok{ title,}
\NormalTok{ frameon }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{ size }\OperatorTok{=}\DecValTok{26}\NormalTok{,}
\NormalTok{ alpha }\OperatorTok{=}\FloatTok{0.9}\NormalTok{,}
\NormalTok{ show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{ return\_fig }\OperatorTok{=}\VariableTok{True}\NormalTok{,}
\NormalTok{ })}
\NormalTok{ out\_file }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\{}\}\NormalTok{ name }\SpecialStringTok{"}
\NormalTok{ save\_plot\_18x18(fig, out\_file)}
\NormalTok{ save\_plot\_18x18(fig, os.path.join(output\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\{}\}\NormalTok{ name }\SpecialStringTok{"}

\ControlFlowTok{if}\NormalTok{ show\_inline:}
\ControlFlowTok{pass}
\NormalTok{ plt.close(fig)}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"Root{-}cell plot saved: "}\SpecialCharTok{\{}\}\NormalTok{ out\_file})}
\ControlFlowTok{return} \NormalTok{ root\_cell}

\CommentTok{\# =====}
\CommentTok{\# STEP 27 {-}{-} run\_trajectory\_runs (function the notebook calls)}
\CommentTok{\# =====}
\CommentTok{\# Runs one or more trajectory parameter sets, saves each result in a separate}
\CommentTok{\# folder, and returns the selected trajectory for downstream steps.}
\KeywordTok{def}\NormalTok{ run\_trajectory\_runs(}
\NormalTok{ adata,}
\NormalTok{ clusters,}
\NormalTok{ root\_cluster,}
\NormalTok{ annotation\_col,}

```

```

\NormalTok{      output\_base\_dir,}
\NormalTok{      runs,}
\NormalTok{      selected\_run}\OperatorTok{=}\VariableTok{None}\NormalTok{,}
\NormalTok{      show\_inline}\OperatorTok{=}\VariableTok{True}\NormalTok{,}
\NormalTok{    }
\NormalTok{  trajectory\_runs }\OperatorTok{=}\NormalTok{ {}}

\ControlFlowTok{for}\NormalTok{ run }\KeywordTok{in}\NormalTok{ runs:}
\NormalTok{  run\_name }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"name"}}
\ControlFlowTok{if}\KeywordTok{not}\NormalTok{ run\_name:}
\ControlFlowTok{raise}\PreprocessorTok{ValueError}\NormalTok{ (}\StringTok{"Each trajectory run needs a name"}
\NormalTok{  run\_dir }\OperatorTok{=}\NormalTok{ os.path.join(output\_base\_dir, }\StringTok{"trajectory"}
\NormalTok{  os.makedirs(run\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{)}

\NormalTok{  nodes }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"nodes"}}
\NormalTok{  sigma }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"sigma"}}
\NormalTok{  ppt\_lambda }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"lambda"}}
\NormalTok{  n\_eigs }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"eigs"}}
\NormalTok{  n\_neighbors }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"n\_neighbors"}}
\NormalTok{  seed }\OperatorTok{=}\NormalTok{ run.get()\StringTok{"seed"}}

\NormalTok{  param\_label }\OperatorTok{=}\NormalTok{ (}
\SpecialStringTok{"nodes="}\SpecialCharTok{\}\NormalTok{nodes}\SpecialCharTok{\}\SpecialStringTok{, sigma="}\SpecialCharTok{\}\NormalTok{sigma}\SpecialCharTok{\}\SpecialStringTok{, ppt\_lambda="}\SpecialCharTok{\}\NormalTok{ppt\_lambda}\SpecialCharTok{\}\NormalTok{, n\_eigs="}\SpecialCharTok{\}\NormalTok{n\_eigs}\SpecialCharTok{\}\NormalTok{, n\_neighbors="}\SpecialCharTok{\}\NormalTok{n\_neighbors}\SpecialCharTok{\}\NormalTok{, seed="}\SpecialCharTok{\}\NormalTok{seed}\SpecialCharTok{\}\SpecialStringTok{)}

\BuiltInTok{print}\NormalTok{ (}\SpecialStringTok{"Running: "}\CharTok{\textbackslash{n}}\SpecialStringTok{Running: }\NormalTok{run\_name}\SpecialStringTok{ }\NormalTok{)}
\BuiltInTok{print}\NormalTok{ (}\SpecialStringTok{"Results folder: "}\CharTok{\textbackslash{n}}\NormalTok{run\_dir}\SpecialStringTok{ }\NormalTok{)}

\NormalTok{  adata\_run }\OperatorTok{=}\NormalTok{ build\_pseudotime\_trajectory(}
\NormalTok{    adata }\OperatorTok{=}\NormalTok{ adata,}
\NormalTok{    clusters }\OperatorTok{=}\NormalTok{ clusters,}
\NormalTok{    root\_cluster }\OperatorTok{=}\NormalTok{ root\_cluster,}
\NormalTok{    annotation\_col }\OperatorTok{=}\NormalTok{ annotation\_col,}
\NormalTok{    nodes }\OperatorTok{=}\NormalTok{ nodes,}
\NormalTok{    sigma }\OperatorTok{=}\NormalTok{ sigma,}
\NormalTok{    ppt\_lambda }\OperatorTok{=}\NormalTok{ ppt\_lambda,}
\NormalTok{    n\_eigs }\OperatorTok{=}\NormalTok{ n\_eigs,}
\NormalTok{    n\_components }\OperatorTok{=}\DecValTok{50}\NormalTok{,}
\NormalTok{    n\_neighbors }\OperatorTok{=}\NormalTok{ n\_neighbors,}
\NormalTok{    seed }\OperatorTok{=}\NormalTok{ seed,}
\NormalTok{  )}

\NormalTok{  params }\OperatorTok{=}\NormalTok{ {}}
\NormalTok{  params.update(}\StringTok{"root\_cluster"}\NormalTok{: root\_cluster)}
\NormalTok{  adata\_run.write\_h5ad(os.path.join(run\_dir, }\StringTok{"adata.h5ad"}
\NormalTok{  pd.Series(params).to\_csv(os.path.join(run\_dir, }\StringTok{"parameters.tsv"}
\NormalTok{  sep=}\StringTok{"\t"}

\NormalTok{  plot\_trajectory\_graphs(}
\NormalTok{    adata }\OperatorTok{=}\NormalTok{ adata\_run,}
\NormalTok{    name }\OperatorTok{=}\NormalTok{ run\_name,}
\NormalTok{    output\_dir }\OperatorTok{=}\NormalTok{ run\_dir,}
\NormalTok{    annotation\_col }\OperatorTok{=}\NormalTok{ annotation\_col,}
\NormalTok{    show\_inline }\OperatorTok{=}\NormalTok{ show\_inline,}
\NormalTok{    param\_label }\OperatorTok{=}\NormalTok{ param\_label,}
\NormalTok{  )}

\NormalTok{  export\_pseudotime\_table(}
\NormalTok{    adata }\OperatorTok{=}\NormalTok{ adata\_run,}
\NormalTok{    name }\OperatorTok{=}\NormalTok{ run\_name,}
\NormalTok{    output\_dir }\OperatorTok{=}\NormalTok{ run\_dir,}

```

```

\NormalTok{      annotation\_col } \OperatorTok{=} \NormalTok{ annotation\_col,}
\NormalTok{    }}

\NormalTok{    plot\_pseudotime\_trajectory(}
\NormalTok{      adata      } \OperatorTok{=} \NormalTok{ adata\_run,}
\NormalTok{      name       } \OperatorTok{=} \NormalTok{ run\_name,}
\NormalTok{      output\_dir } \OperatorTok{=} \NormalTok{ run\_dir,}
\NormalTok{      annotation\_col } \OperatorTok{=} \NormalTok{ annotation\_col,}
\NormalTok{      show\_inline } \OperatorTok{=} \NormalTok{ show\_inline,}
\NormalTok{      param\_label } \OperatorTok{=} \NormalTok{ param\_label,}
\NormalTok{    )}

\NormalTok{    plot\_root\_cell(}
\NormalTok{      adata      } \OperatorTok{=} \NormalTok{ adata\_run,}
\NormalTok{      name       } \OperatorTok{=} \NormalTok{ run\_name,}
\NormalTok{      output\_dir } \OperatorTok{=} \NormalTok{ run\_dir,}
\NormalTok{      show\_inline } \OperatorTok{=} \NormalTok{ show\_inline,}
\NormalTok{      param\_label } \OperatorTok{=} \NormalTok{ param\_label,}
\NormalTok{    )}

\NormalTok{    trajectory\_runs[run\_name] } \OperatorTok{=} \NormalTok{ \{ }
\StringTok{"adata"} \NormalTok{ : adata\_run,}
\StringTok{"output\_dir"} \NormalTok{ : run\_dir,}
\StringTok{"params"} \NormalTok{ : params,}
\NormalTok{    \} }

\ControlFlowTok{if} \NormalTok{ selected\_run } \KeywordTok{is} \VariableTok{None} \NormalTok{ :}
\NormalTok{    selected\_run } \OperatorTok{=} \NormalTok{ runs } \OperatorTok{{-}} \DecValTok{1} \NormalTok{ .get( ) \StringTok{"selected\_run"} }

\ControlFlowTok{if} \NormalTok{ selected\_run } \KeywordTok{not} \KeywordTok{in} \NormalTok{ trajectory\_runs :}
\NormalTok{    available } \OperatorTok{=} \BuiltInTok{list} \NormalTok{ (trajectory\_runs) }
\ControlFlowTok{raise} \PreprocessorTok{ValueError} \NormalTok{ (} \StringTok{"selected\_run \textquotesingle"}

\NormalTok{    selected } \OperatorTok{=} \NormalTok{ trajectory\_runs[selected\_run] }
\BuiltInTok{print} \NormalTok{ (} \StringTok{"STEP 27 COMPLETE"}
\BuiltInTok{print} \NormalTok{ (} \StringTok{"Selected run: "} \NormalTok{ selected\_run } \StringTok{"Selected folder: "} \NormalTok{ selected\_run } \StringTok{"output\_dir"}

\ControlFlowTok{return} \NormalTok{ selected\_run } \StringTok{"adata"} \NormalTok{ , selected\_run } \StringTok{"output\_dir"}

\CommentTok{\# =====}
\CommentTok{\# STEP 28 {-}{-} Plot genes on trajectory}
\CommentTok{\# =====}
\CommentTok{\# No wrapper function {-}{-} this step is now just 3 lines of plain scanpy in the}
\CommentTok{\# notebook cell itself (gene\_plots/ folder + figdir + sc.pl.draw\_graph).}
\CommentTok{\# Kept out of this file on purpose so the notebook cell is fully self-contained.}

\KeywordTok{def} \NormalTok{ strip\_segment\_frames(ax):}
\NormalTok{    xlim } \OperatorTok{=} \NormalTok{ ax.get\_xlim() }
\NormalTok{    full\_width } \OperatorTok{=} \NormalTok{ xlim[0] } \DecValTok{1} \NormalTok{ } \OperatorTok{{-}} \NormalTok{ x }
\ControlFlowTok{for} \NormalTok{ p } \KeywordTok{in} \BuiltInTok{list} \NormalTok{ (ax.patches):}
\ControlFlowTok{if} \NormalTok{ BuiltInTok{isinstance} } \NormalTok{ (p, mpatches.Rectangle):}
\NormalTok{    ec } \OperatorTok{=} \NormalTok{ p.get\_edgcolor() }
\NormalTok{    is\_black } \OperatorTok{=} \NormalTok{ ec[:3] } \DecValTok{3} \NormalTok{ } \OperatorTok{=} \NormalTok{ is\_black }
\NormalTok{    is\_unfilled } \OperatorTok{=} \NormalTok{ (} \KeywordTok{not} \NormalTok{ p.get\_fill( ) ) } \KeywordTok{and} \NormalTok{ p.get\_fill( ) } \KeywordTok{is} \NormalTok{ is\_unfilled } \KeywordTok{and} \NormalTok{ is\_black }
\NormalTok{    p.remove( )}

```

```

\CommentTok{\# Trends plots get their own canvas (24x18, wider than tall) instead of the}
\CommentTok{\# generic PLOT\FIGSIZE (18x18) used by other plot types {-}{-} the dendrogram}
\CommentTok{\# panel looks cramped at a square 18x18 once the tree is a single segment.}
\KeywordTok{def}\NormalTok{ save\_trends\_plot(axes\_list, png\_path, pdf\_path):}
    \ControlFlowTok{for}\NormalTok{ ax }\KeywordTok{in}\NormalTok{ axes\_list:}
\NormalTok{    strip\_segment\_frames(ax)}
\NormalTok{    fig }\OperatorTok{=}\NormalTok{ axes\_list[0]\NormalTok{.get\_figure()}
\NormalTok{    fig.set\_size\_inches(}\DecValTok{24}\NormalTok{, }\DecValTok{18}\NormalTok{, forward)\OperatorTok{=}\NormalTok{V}
\NormalTok{    fig.subplots\_adjust(left)\OperatorTok{=}\FloatTok{0.04}\NormalTok{, right)\OperatorTok{=}\FloatTok{0.8}
\NormalTok{    fig.savefig(png\_path, dpi)\OperatorTok{=}\NormalTok{PLOT\_DPI, facecolor}\OperatorTok{=}\StringTok{"wh}
\NormalTok{    fig.savefig(pdf\_path, dpi)\OperatorTok{=}\NormalTok{PLOT\_DPI, facecolor}\OperatorTok{=}\StringTok{"wh}
\NormalTok{    plt.close(fig)}
\NormalTok{    plt.close(fig)}

\KeywordTok{def}\NormalTok{ run\_step29\_gene\_trends(}
\NormalTok{    adata,}
\NormalTok{    run\_dir,}
\NormalTok{    name }\OperatorTok{=}\StringTok{"trajectory"}\NormalTok{,}
\NormalTok{    custom\_genes }\OperatorTok{=}\VariableTok{None}\NormalTok{,}
\NormalTok{    top\_n }\OperatorTok{=}\DecValTok{50}\NormalTok{,}
\NormalTok{    ordering }\OperatorTok{=}\StringTok{"max"}\NormalTok{,}
\NormalTok{    n\_jobs }\OperatorTok{=}\DecValTok{4}\NormalTok{,}
\NormalTok{    spline\_df }\OperatorTok{=}\DecValTok{5}\NormalTok{,}
\NormalTok{    A\_cut }\OperatorTok{=}\FloatTok{0.3}\NormalTok{,}
\NormalTok{    p\_val\_cut }\OperatorTok{=}\FloatTok{0.001}\NormalTok{,}
\NormalTok{):}
\NormalTok{    output\_dir }\OperatorTok{=}\NormalTok{ os.path.join(run\_dir, }\StringTok{"gene\_trends"}\NormalTok{)}
\NormalTok{    os.makedirs(output\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{)}

    \ControlFlowTok{if}\NormalTok{ adata.X }\KeywordTok{is}\VariableTok{None}\KeywordTok{and}\StringTok{"logcounts"}
\NormalTok{    adata.X }\OperatorTok{=}\NormalTok{ adata.layers[}\StringTok{"logcounts"}\NormalTok{]}

    \CommentTok{\# spline\_df: degrees of freedom of the per{-}segment GAM smoothing spline}
    \CommentTok{\# used by test\_association. Lower it (e.g. 3) if you hit "A term has}
    \CommentTok{\# fewer unique covariate combinations than specified maximum degrees of}
    \CommentTok{\# freedom" {-}{-} that means some tree segment has too few distinct pseudotime}
    \CommentTok{\# values for the default of 5.}
    \CommentTok{\#}
    \CommentTok{\# A\_cut / p\_val\_cut: significance is redefined from scFates\textquotesingle{s own}
    \CommentTok{\# combined fdr/A/st cutoffs (very conservative on small datasets) to a}
    \CommentTok{\# lower amplitude cutoff plus the raw (non{-}FDR{-}corrected) p{-}value {-}{-} the}
    \CommentTok{\# global/whole{-}tree version of the technique used per{-}branch in pse.py.}
\NormalTok{    scf.tl.dendrogram(adata)}
\NormalTok{    scf.tl.test\_association(adata, n\_jobs)\OperatorTok{=}\NormalTok{n\_jobs, spline\_df)\OperatorTok{=}\NormalTok{N}
\NormalTok{    adata.var[}\StringTok{"signi"}\NormalTok{] }\OperatorTok{=}\NormalTok{ adata.var[}\StringTok{"p\_val"}\NormalTok{]}
\NormalTok{    significant\_genes }\OperatorTok{=}\NormalTok{ adata.var\_names[adata.var[}\StringTok{"signi"}\NormalTok{]}

    \CommentTok{\# Custom genes are force{-}included in the fit even if not statistically}
    \CommentTok{\# significant, so a gene you explicitly asked for never silently}
    \CommentTok{\# disappears from either heatmap.}
\NormalTok{    custom\_genes }\OperatorTok{=}\NormalTok{ custom\_genes }\KeywordTok{or}\NormalTok{ []}
\NormalTok{    genes\_present }\OperatorTok{=}\NormalTok{ [g }\ControlFlowTok{for}\NormalTok{ g }\KeywordTok{in}\NormalTok{ }
\NormalTok{    missing }\OperatorTok{=}\NormalTok{ }\BuiltInTok{sorted}\NormalTok{(\}\BuiltInTok{set}\NormalTok{(\custom\_gene}
    \ControlFlowTok{if}\NormalTok{ missing:}
        \BuiltInTok{print}\NormalTok{(\}\SpecialStringTok{f"custom selection: genes ignored (not found in adata): }\Spe}
\NormalTok{    fit\_features }\OperatorTok{=}\NormalTok{ }\BuiltInTok{list}\NormalTok{(\}\BuiltInTok{dict}\NormalTok{.fromkeys(signi}

    \CommentTok{\# Fit on a copy so the caller\textquotesingle{s adata is never subsetted. Both heatmaps}
    \CommentTok{\# below reuse this single fit {-}{-} rows = significant + custom genes, only}
    \CommentTok{\# the highlighted/labeled subset differs between the two plots.}

```

```

\NormalTok{      adata\_fitted }\OperatorTok{=}\NormalTok{ adata.copy()}
\NormalTok{      scf.tl.fit(adata\_fitted, features)\OperatorTok{=}\NormalTok{fit\_features, n\_jobs}\OperatorTok{=}\NormalTok{ }

\NormalTok{      top\_genes }\OperatorTok{=}\NormalTok{ adata\_fitted.var.sort\_values()\StringTok{"A"}\NormalTok{, asce}
\NormalTok{      axes\_list }\OperatorTok{=}\NormalTok{ scf.pl.trends(}
\NormalTok{          adata\_fitted,}
\NormalTok{          highlight\_features }\OperatorTok{=}\NormalTok{ top\_genes,}
\NormalTok{          style }\OperatorTok{=}\NormalTok{ \StringTok{"italic"}\NormalTok{,}
\NormalTok{          add\_outline }\OperatorTok{=}\NormalTok{ \VariableTok{True}\NormalTok{,}
\NormalTok{          basis }\OperatorTok{=}\NormalTok{ \StringTok{"dendro"}\NormalTok{,}
\NormalTok{          show\_segs }\OperatorTok{=}\NormalTok{ \VariableTok{False}\NormalTok{,}
\NormalTok{          fontsize }\OperatorTok{=}\NormalTok{ \DecValTok{18}\NormalTok{,}
\NormalTok{          ordering }\OperatorTok{=}\NormalTok{ ordering,}
\NormalTok{          show }\OperatorTok{=}\NormalTok{ \VariableTok{False}\NormalTok{,}
\NormalTok{          title }\OperatorTok{=}\NormalTok{ \SpecialStringTok{f"}\SpecialCharTok{\{}\NormalTok{name}\SpecialCharTok{}}
\NormalTok{      )}
\NormalTok{      top\_png }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\SpecialStringTok{f"step29\_gene\_trend}
\NormalTok{      top\_pdf }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\SpecialStringTok{f"step29\_gene\_trend}
\NormalTok{      save\_trends\_plot(axes\_list, top\_png, top\_pdf)}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"[OK] Top-}\SpecialCharTok{\{}\NormalTok{top\_n}\SpecialCharTok{}}

\NormalTok{      custom\_png }\OperatorTok{=}\NormalTok{ \VariableTok{None}}
\ControlFlowTok{if}\NormalTok{ genes\_present:}
\NormalTok{      axes\_list }\OperatorTok{=}\NormalTok{ scf.pl.trends(}
\NormalTok{          adata\_fitted,}
\NormalTok{          highlight\_features }\OperatorTok{=}\NormalTok{ genes\_present,}
\NormalTok{          style }\OperatorTok{=}\NormalTok{ \StringTok{"italic"}\NormalTok{,}
\NormalTok{          add\_outline }\OperatorTok{=}\NormalTok{ \VariableTok{True}\NormalTok{,}
\NormalTok{          basis }\OperatorTok{=}\NormalTok{ \StringTok{"dendro"}\NormalTok{,}
\NormalTok{          show\_segs }\OperatorTok{=}\NormalTok{ \VariableTok{False}\NormalTok{,}
\NormalTok{          fontsize }\OperatorTok{=}\NormalTok{ \DecValTok{18}\NormalTok{,}
\NormalTok{          ordering }\OperatorTok{=}\NormalTok{ ordering,}
\NormalTok{          show }\OperatorTok{=}\NormalTok{ \VariableTok{False}\NormalTok{,}
\NormalTok{          title }\OperatorTok{=}\NormalTok{ \SpecialStringTok{f"}\SpecialCharTok{\{}\NormalTok{name}\SpecialCharTok{}}
\NormalTok{      )}
\NormalTok{      custom\_png }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\StringTok{"step29\_gene\_trends}
\NormalTok{      custom\_pdf }\OperatorTok{=}\NormalTok{ os.path.join(output\_dir, }\StringTok{"step29\_gene\_trends}
\NormalTok{      save\_trends\_plot(axes\_list, custom\_png, custom\_pdf)}
\BuiltInTok{print}\NormalTok{({}\SpecialStringTok{f"[OK] Custom selection gene trends saved: }\SpecialCharTok{}}

\BuiltInTok{print}\NormalTok{({}\StringTok{"STEP 29 COMPLETE: gene trends saved"}\NormalTok{)}
\ControlFlowTok{return}\NormalTok{ top\_png, custom\_png}

```

9 capitulo3_pseudotime.py

Script principal de capítulo 3 (versión .py plana del notebook capitulo3_pseudotime.ipynb, funcionalmente equivalente y más legible en texto).

```
\CommentTok{\# =====}
\CommentTok{\# capitulo3\_pseudotime.py {-} Pseudotime trajectory analysis (Part 3)}
\CommentTok{\# =====}
\CommentTok{\# Requires: pbmc\_harmony\_curated.h5ad generated by capitulo1\_single\_cell.R}
\CommentTok{\#}
\CommentTok{\# How to use:}
\CommentTok{\# 1. Run sections in order.}
\CommentTok{\# 2. Edit only the small parameter blocks marked "Edit here".}
\CommentTok{\# 3. All plots are saved as 18 x 18.}
\CommentTok{\# =====}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 24 {-} SETUP}
\CommentTok{\# =====}
\CommentTok{\# Sets paths and loads Python libraries/functions.}
\CommentTok{\# Change PIPELINE\_DIR and DATA\_DIR only if your folder structure is different.}

\ImportTok{import}\NormalTok{ os}

\NormalTok{PIPELINE\_DIR }\OperatorTok{=} \StringTok{"/workspace/workflow\_nuevo"}
\NormalTok{DATA\_DIR }\OperatorTok{=} \StringTok{"/workspace/."}
\NormalTok{base\_dir }\OperatorTok{=} \NormalTok{os.path.join(DATA\_DIR, }\StringTok{"resultados"}\NormalTok{)}

\BuiltInTok{exec}\NormalTok{({}\BuiltInTok{open}\NormalTok{os.path.join(PIPELINE\_DIR, }\StringTok{"load\_libraries\_p"}
\BuiltInTok{exec}\NormalTok{({}\BuiltInTok{open}\NormalTok{os.path.join(PIPELINE\_DIR, }\StringTok{"ScRNA\_Pseudotime"}

\NormalTok{dir\_pseudotime }\OperatorTok{=} \NormalTok{os.path.join(base\_dir, }\StringTok{"09\_pseudotime"}\NormalTok{)}
\NormalTok{os.makedirs(dir\_pseudotime, exist\_ok)\OperatorTok{=} \VariableTok{True}\NormalTok{)}

\BuiltInTok{print}\NormalTok{({}\StringTok{"SECTION 24 COMPLETE: Setup done"}\NormalTok{)}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 25 {-} LOAD CURATED OBJECT}
\CommentTok{\# =====}
\CommentTok{\# Loads the AnnData object exported from capitulo1.}
\CommentTok{\#}
\CommentTok{\# What this section does:}
\CommentTok{\# 1. Reads the .h5ad object with cells, genes and metadata.}
\CommentTok{\# 2. Converts Seurat{-}style UMAP/PCA/HARMONY coordinates to scanpy names.}
\CommentTok{\# 3. Prints available cell types for Section 26.}
\CommentTok{\# 4. Saves overview UMAP plots.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\# INPUT\_H5AD = curated .h5ad from capitulo1}
\CommentTok{\# ANNOTATION\_COL = metadata column with final cell{-}type labels}
\CommentTok{\# N\_JOBS = CPU cores for parallel Python steps}

\NormalTok{INPUT\_H5AD }\OperatorTok{=} \NormalTok{os.path.join({}\StringTok{"/workspace/resultados/objects/pbmc\_h"}
\NormalTok{ANNOTATION\_COL }\OperatorTok{=} \StringTok{"celltype\_curated"}
\NormalTok{N\_JOBS }\OperatorTok{=} \DecValTok{4}

\NormalTok{adata, N\_JOBS }\OperatorTok{=} \NormalTok{load\_curated\_object({}
\NormalTok{input\_h5ad }\OperatorTok{=} \NormalTok{INPUT\_H5AD,}
```



```

\NormalTok{ dir\_pseudotime }\OperatorTok{=}\NormalTok{ dir\_pseudotime,}
\NormalTok{ annotation\_col }\OperatorTok{=}\NormalTok{ ANNOTATION\_COL,}
\NormalTok{ n\_jobs }\OperatorTok{=}\NormalTok{ N\_JOBS,}
\NormalTok{}}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 26 {-} CELL TYPE SELECTION}
\CommentTok{\# =====}
\CommentTok{\# Selects the cell types used to build the trajectory.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\# Copy names exactly from the Section 25 list.}

\NormalTok{TRAJECTORY\_CLUSTERS }\OperatorTok{=}\NormalTok{ []\StringTok{"Pavement Cell"}\NormalTok{, }\StringTok{"Stomatal Line"}\NormalTok{}}

\NormalTok{adata\_sub }\OperatorTok{=}\NormalTok{ preview\_trajectory\_selection()}
\NormalTok{ adata }\OperatorTok{=}\NormalTok{ adata,}
\NormalTok{ clusters }\OperatorTok{=}\NormalTok{ TRAJECTORY\_CLUSTERS,}
\NormalTok{ annotation\_col }\OperatorTok{=}\NormalTok{ ANNOTATION\_COL,}
\NormalTok{ dir\_pseudotime }\OperatorTok{=}\NormalTok{ dir\_pseudotime,}
\NormalTok{}}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 27 {-} TRAJECTORY INFERENCE}
\CommentTok{\# =====}
\CommentTok{\# Builds one or more trajectory runs using the selected cell types.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\# ROOT\_CLUSTER = biological start point where pseudotime = 0}
\CommentTok{\# Add more trajectory\_run(...) lines to test more parameter sets.}
\CommentTok{\#}
\CommentTok{\# Parameter meaning:}
\CommentTok{\# nodes = more nodes gives a finer tree}
\CommentTok{\# sigma = lower values follow the cells more tightly}
\CommentTok{\# lambda\_value = higher values make a simpler tree}
\CommentTok{\# eigs = diffusion{-}map dimensions used for trajectory geometry}
\CommentTok{\# seed = reproducibility}

\NormalTok{ROOT\_CLUSTER }\OperatorTok{=}\StringTok{"Stomatal Line"}

\NormalTok{TRAJECTORY\_RUNS }\OperatorTok{=}\NormalTok{ []}
\NormalTok{ trajectory\_run(nodes)\OperatorTok{=}\DecValTok{25}\NormalTok{, sigma}\OperatorTok{=}\FloatTok{0.5}\NormalTok{,}
\NormalTok{}}

\NormalTok{adata\_traj, selected\_trajectory\_dir, trajectory\_runs }\OperatorTok{=}\NormalTok{ run\_trajectory\_runs(}
\NormalTok{ adata }\OperatorTok{=}\NormalTok{ adata,}
\NormalTok{ clusters }\OperatorTok{=}\NormalTok{ TRAJECTORY\_CLUSTERS,}
\NormalTok{ root\_cluster }\OperatorTok{=}\NormalTok{ ROOT\_CLUSTER,}
\NormalTok{ annotation\_col }\OperatorTok{=}\NormalTok{ ANNOTATION\_COL,}
\NormalTok{ output\_base\_dir }\OperatorTok{=}\NormalTok{ dir\_pseudotime,}
\NormalTok{ runs }\OperatorTok{=}\NormalTok{ TRAJECTORY\_RUNS,}
\NormalTok{}}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 28 {-} PLOT GENES ON TRAJECTORY}

```

```

\CommentTok{\# =====}
\CommentTok{\# Plots the selected genes on the trajectory layout.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\# Write the gene IDs you want to visualize.}

\NormalTok{GENES }\OperatorTok{=}\NormalTok{[ ]}\StringTok{"AT1G22690"}\NormalTok{, }\StringTok{"AT4G04890"}\NormalTok{
\NormalTok{name }\OperatorTok{=}\NormalTok{os.path.basename(selected\_trajectory\_dir)}

\ControlFlowTok{if}\NormalTok{adata\_traj.X }\KeywordTok{is} \VariableTok{None} \KeywordTok{and} \StringTok{"logcount"}
\NormalTok{adata\_traj.X }\OperatorTok{=}\NormalTok{adata\_traj.layers[]}\StringTok{"logcounts"}\NormalTok{[]}}

\NormalTok{gene\_plots\_dir }\OperatorTok{=}\NormalTok{os.path.join(selected\_trajectory\_dir, }\StringTok{"gene\_plo"}
\NormalTok{os.makedirs(gene\_plots\_dir, exist\_ok)\OperatorTok{=}\VariableTok{True}\NormalTok{}}

\NormalTok{fig }\OperatorTok{=}\NormalTok{sc.pl.draw\_graph(}
\NormalTok{adata\_traj,}
\NormalTok{color }\OperatorTok{=}\NormalTok{GENES,}
\NormalTok{title }\OperatorTok{=}\NormalTok{GENES,}
\NormalTok{show }\OperatorTok{=}\VariableTok{False}\NormalTok{,}
\NormalTok{return\_fig }\OperatorTok{=}\VariableTok{True}\NormalTok{,}
\NormalTok{)}
\NormalTok{save\_plot\_18x18(fig, os.path.join(gene\_plots\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\}\NormalTok{n"}
\NormalTok{save\_plot\_18x18(fig, os.path.join(gene\_plots\_dir, }\SpecialStringTok{f"}\SpecialCharTok{\}\NormalTok{n"}
\NormalTok{plt.close(fig)}

\BuiltInTok{print}\NormalTok{({}\StringTok{"SECTION 28 COMPLETE: Gene trajectory plots saved"}\NormalTok{)}}

\CommentTok{\# \%}%}
\CommentTok{\# =====}
\CommentTok{\# SECTION 29 {-} GENE TRENDS}
\CommentTok{\# =====}
\CommentTok{\# Saves two heatmaps:}
\CommentTok{\# 1. Top variable genes along the trajectory.}
\CommentTok{\# 2. Custom genes highlighted by the user.}
\CommentTok{\#}
\CommentTok{\# Edit here:}
\CommentTok{\# TOP\_N = number of top genes to highlight in the first plot}
\CommentTok{\# HIGHLIGHT\_GENES = genes of interest to highlight in the second plot}
\CommentTok{\# ORDERING = "max", "pearson", or "quantile"}

\NormalTok{TOP\_N }\OperatorTok{=}\NormalTok{DecValTok{10}}
\NormalTok{HIGHLIGHT\_GENES }\OperatorTok{=}\NormalTok{[ ]}\StringTok{"AT1G22690"}\NormalTok{, }\StringTok{"AT4G04890"}\NormalTok{
\NormalTok{ORDERING }\OperatorTok{=}\NormalTok{"max"}
\CommentTok{\# Lower this (e.g. 3) if you hit "fewer unique covariate combinations than"}
\CommentTok{\# specified maximum degrees of freedom" during test\_association.}
\NormalTok{SPLINE\_DF }\OperatorTok{=}\NormalTok{DecValTok{5}}

\NormalTok{top\_path, highlight\_path }\OperatorTok{=}\NormalTok{run\_step29\_gene\_trends(}
\NormalTok{adata }\OperatorTok{=}\NormalTok{adata\_traj,}
\NormalTok{run\_dir }\OperatorTok{=}\NormalTok{selected\_trajectory\_dir,}
\NormalTok{name }\OperatorTok{=}\NormalTok{os.path.basename(selected\_trajectory\_dir),}
\NormalTok{custom\_genes }\OperatorTok{=}\NormalTok{HIGHLIGHT\_GENES,}
\NormalTok{top\_n }\OperatorTok{=}\NormalTok{TOP\_N,}
\NormalTok{ordering }\OperatorTok{=}\NormalTok{ORDERING,}
\NormalTok{n\_jobs }\OperatorTok{=}\NormalTok{N\_JOBS,}
\NormalTok{spline\_df }\OperatorTok{=}\NormalTok{SPLINE\_DF,}
\NormalTok{)}

\BuiltInTok{print}\NormalTok{({}\StringTok{"SECTION 29 COMPLETE: Gene trend plots saved"}\NormalTok{)}}

```

```
\CommentTok{\# \%}%}  
\CommentTok{\# =====}  
\CommentTok{\# CAPITULO 3 COMPLETE}  
\CommentTok{\# =====}  
\BuiltInTok{print}\NormalTok{({}\StringTok{"CAPITULO 3 COMPLETE"}\NormalTok{{})}
```